HAML: Hierarchical Attractor Machine Learning with Bidirectional Coupling for Supervised Classification

A Preliminary Investigation

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Abstract

We introduce HAML (Hierarchical Attractor Machine Learning), an exploratory supervised classification framework in which classification emerges from the convergence of a coupled dynamical system toward stable basins of attraction across multiple hierarchical levels. Each level operates in a PCA-decomposed subspace of decreasing dimension; levels are coupled through bottom-up (error-prediction) and top-down (context-prediction) feedback flows inspired by Predictive Coding.

We present four theoretical contributions: (1) a dissipative-convergence argument via LaSalle's invariance principle (Proposition 4.1); (2) a dimension-independent local convergence-rate result at fixed points under Gaussian potentials with dimension-scaled kernel initialisation (Proposition 4.6); (3) a hierarchical-decomposition argument giving exponential gains in attractor count on problems of high topological complexity, with a sub-linear Rademacher bound in depth (Propositions 4.10–4.13); (4) an exact correspondence with hierarchical Predictive Coding MAP inference under conditions M1–M3 (Theorem 4.15).

We further introduce a three-phase progressive training strategy to stabilise gradient learning of coupled dynamics, and a level-aware collapse-recovery mechanism for inter-level representation collapse.

Empirical results (honest summary). The empirical study consists of six campaigns; we summarise them with their actual statistical status rather than headline numbers:

- On a MNIST subset (1500 train / 500 test, 5 epochs), gradient training raises test accuracy from 45.6% (k-means-only initialisation) to 75.8%. On this short protocol, bidirectional coupling provides **no measurable gain** over the independent-levels variant (75.8% vs. 75.8%), which we discuss explicitly rather than hide.

- On alternating concentric rings (5 seeds, 42–46), coupled HAML reaches $71.20\% \pm 6.43$ vs. $67.30\% \pm 4.37$ for the independent variant ($\Delta = +3.90$ pts, *not statistically significant* at this sample size). The qualitative recovery of non-convex decision boundaries by the coupled model is nonetheless visible in the learned geometry.
- On noisy moons (single seed), the coupled gain peaks at intermediate noise (+6.27 pts at noise 0.20). This is reported as a single-seed observation consistent with the theoretical prediction; multi-seed validation is required.
- On Fashion-MNIST (784D, 10 classes, 5000 train / 1000 test, single seed, 10 epochs), test accuracy reaches 80.1%. We do *not* claim this is competitive with standard baselines (a logistic regression typically reaches ~84% and a small CNN > 90% on this dataset); the figure demonstrates that the framework trains stably in 784D rather than that it outperforms standard methods.
- A controlled *reimplementation* of SA-nODE [5] (single seed, not using the authors' official code) is dominated by HAML across noise levels; we emphasise the preliminary nature of this comparison.

Status of this work. This is an exploratory preprint from an independent researcher. The theoretical and architectural contributions are intended as a coherent proposal; the empirical evidence is preliminary and explicitly under-powered (single seeds on most real-data benchmarks, no comparison to strong neural baselines, internal SA-nODE reimplementation). The repository contains the full code and configuration files required to reproduce every reported number; we welcome external replication.

Keywords: Supervised learning, attractor dynamics, hierarchical classification, neural ODE, predictive coding, energy-based models, dynamical systems.

1 Introduction

Supervised classification traditionally relies on static function approximation through neural networks or kernel methods. We explore an alternative perspective: *classification as dynamic convergence.* Rather than computing a forward mapping $f : \mathcal{X} \to \mathcal{Y}$ in one pass, the model learns to position attractors (stable equilibrium points) in hierarchical latent subspaces so that unseen samples converge to basins corresponding to their true class under coupled gradient dynamics.

This approach builds on four established research strands:

- **Dynamical systems theory**: stable manifolds and attractor landscapes provide geometrically interpretable decision boundaries.
- **Predictive coding** [2, 6]: top-down predictions correct bottom-up prediction errors across a generative hierarchy.
- **Energy-based models**: classification emerges from minimisation of a learned potential energy.
- **Modern Hopfield networks** [7]: continuous attractors replace discrete states for greater expressivity.

1.1 Scope and Disclaimer

We state upfront what this paper does and does not claim, to set expectations correctly:

What this paper claims. (i) A coherent architectural proposal combining hierarchical PCA-decomposed state spaces, learned continuous attractors, and bidirectional Predictive-Coding-style coupling. (ii) Four theoretical results of the form stated in the abstract, with their conditions made explicit. (iii) A practical training recipe (three-phase progressive coupling, level-aware collapse recovery) that empirically stabilises gradient-based optimisation of the coupled dynamics. (iv) Preliminary empirical evidence that the framework trains stably and produces interpretable attractor geometries.

What this paper does *not* claim. (i) State-of-the-art results on any benchmark. (ii) A definitive empirical advantage of bidirectional coupling: the evidence is mixed (positive on rings and noisy moons at intermediate noise, neutral on MNIST). (iii) Superiority over SA-nODE based on the official code of Marino et al. [5] (we use a controlled internal reimplementation). (iv) Multi-seed certification on real high-dimensional data: Fashion-MNIST and MNIST results are single-seed for compute reasons, and we treat them as such.

1.2 Motivation and Related Work

Recent advances in neural differential equations [1] enable end-to-end learning of continuous dynamical systems. SA-nODE [5] introduced a flat attractor model with binary fixed attractors in a single space. We extend this idea along four axes:

1. **Hierarchical decomposition**: multiple coupled levels via PCA subspaces instead of a single ambient space.
2. **Continuous learned attractors**: gradient-optimised positions $\boldsymbol{\mu}$, widths $\boldsymbol{\sigma}$, repulsion radii $\boldsymbol{\rho}$, and weights w.

3. **Bidirectional coupling**: explicit bottom-up error correction and top-down context flow at every level.

4. **Parameterised potentials**: Gaussian kernels with dimension-dependent initialisation to prevent kernel saturation in high dimensions.

Table 1 summarises the architectural differences.

Dimension	**SA-nODE**	**HAML**
State space	Full input dim.	Hierarchy of decreasing subspaces
Attractors	Binary eigenvectors $\pm a$	Continuous, gradient-optimised
Potential	Fixed double-well	Parametric Gaussian mixture
Structure	Single dynamic level	L bidirectionally coupled levels
Coupling	None	Bottom-up + top-down
Local conv. rate at FP	Dimension-dependent	Dim.-independent (Prop. 4.6)
Theoretical anchor	Statistical mechanics	Dyn. systems + Predictive Coding
PC/MAP equivalence	Not established	Exact under M1–M3 (Thm. 4.15)

Table 1: Architectural comparison with SA-nODE [5].

1.3 Contributions

1. **Architecture**: a hierarchical bidirectional attractor system with gradient-learned continuous attractors for supervised classification.

2. **Four theoretical results**: dissipative convergence (Proposition 4.1), dimension-independent local rate at fixed points (Proposition 4.6), hierarchical expressivity gain (Propositions 4.10–4.13), and exact MAP/Predictive Coding correspondence under M1–M3 (Theorem 4.15).

3. **Preliminary empirical study**: 6 campaigns across synthetic and real datasets, with multi-seed runs where compute allowed.

4. **Practical mechanisms**: three-phase progressive training and level-aware collapse recovery.

5. **Open reproducibility**: full source code at `https://github.com/ikobootloader/HIERARCHICAL-ATTRACTOR-MACHINE-LEARNING`.

2 Model Architecture and Dynamics

2.1 Hierarchical State Space

Let $\mathcal{X} \subset \mathbb{R}^d$ be the input space. We define a tower of linear projections:

$$\mathcal{X} = \mathcal{H}_0 \xrightarrow{\pi_1} \mathcal{H}_1 \xrightarrow{\pi_2} \cdots \xrightarrow{\pi_L} \mathcal{H}_L \tag{1}$$

where $\mathcal{H}_l \subset \mathbb{R}^{d_l}$ with $d_0 > d_1 > \cdots > d_L$.

Projections $\pi_l : \mathcal{H}_{l-1} \to \mathcal{H}_l$ are initialised by PCA (capturing maximum variance) and are optionally fine-tuned during training. Their Moore–Penrose pseudo-inverses $\pi_l^\dagger : \mathcal{H}_l \to \mathcal{H}_{l-1}$ serve as upward decoders.

Remark 2.1 (Contraction property)**.** *PCA projections normalised to unit spectral norm satisfy $\|\Pi_l\|_2 \le \sqrt{d_{l+1}/d_l} < 1$. This contraction is used in the convergence analysis (Section 4.1).*

Initial conditions: $\mathbf{x}^{(l)}(0) = \pi_l \circ \cdots \circ \pi_1(\mathbf{x}_{\text{input}})$.

2.2 Attractors and Energy Potential

At level l, we maintain $K \times M$ learnable attractors $\{\boldsymbol{\mu}_{c,m}^{(l)}\}_{c\in[K],m\in[M]}$ with associated parameters $(\sigma_{c,m}^{(l)}, \rho_{c,m}^{(l)}, w_{c,m}^{(l)})$. The energy potential is a parametric attractor–repulsor mixture:

$$E^{(l)}(\mathbf{x}) = -\sum_{c=1}^{K}\sum_{m=1}^{M} w_{c,m}^{(l)} \exp\left(-\frac{\|\mathbf{x}-\boldsymbol{\mu}_{c,m}^{(l)}\|^2}{2(\sigma_{c,m}^{(l)})^2}\right) + \lambda \sum_{c\neq c'}\sum_{m,m'} w_{c',m'}^{(l)} \exp\left(-\frac{\|\mathbf{x}-\boldsymbol{\mu}_{c',m'}^{(l)}\|^2}{2(\rho_{c',m'}^{(l)})^2}\right) \tag{2}$$

Bayesian interpretation (anticipating Section 4.4): without the repulsive terms ($\lambda = 0$), $-E^{(l)}$ is proportional to the log of a Gaussian mixture, i.e. the log-prior $\log p(\mathbf{x}^{(l)})$ in the implicit generative model. Repulsive terms correspond to a discriminative contrastive prior [4].

2.3 Coupled Dynamics

At each level $l \in \{0, \ldots, L\}$, the state evolves under the coupled ODE:

$$\dot{\mathbf{x}}^{(l)}(t) = \mathbf{F}_{\text{intra}}^{(l)}(\mathbf{x}^{(l)}) + \mathbf{F}_{\text{bu}}^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l-1)}) + \mathbf{F}_{\text{td}}^{(l)}(\mathbf{x}^{(l)}, \mathbf{x}^{(l+1)}) - \gamma \mathbf{x}^{(l)} \tag{3}$$

where $\gamma > 0$ is a viscous damping coefficient ensuring energy dissipation.

Intra-level force (local attractors and repulsors):

$$\mathbf{F}_{\text{intra}}^{(l)} = -\nabla_{\mathbf{x}} E^{(l)}(\mathbf{x}^{(l)}) \tag{4}$$

Bottom-up error-correction force:

$$\mathbf{F}_{\text{bu}}^{(l)} = \alpha_{\text{bu}} \cdot \pi_l \left(\mathbf{x}^{(l-1)} - \pi_l^\dagger(\mathbf{x}^{(l)})\right) \tag{5}$$

This is the *prediction error* signal of Predictive Coding [6]: the difference between the actual lower-level state and the reconstruction predicted by the current level.

Top-down prediction force:

$$\mathbf{F}_{\text{td}}^{(l)} = \alpha_{\text{td}} \cdot \left(\pi_l^\dagger\left(\mathbf{x}^{(l+1)}\right) - \mathbf{x}^{(l)}\right) \tag{6}$$

This is the recall force toward the representation predicted by the higher level — the *top-down generation* in the Predictive Coding sense.

Boundary conditions: $\mathbf{F}_{\text{bu}}^{(0)} = 0$, $\mathbf{F}_{\text{td}}^{(L)} = 0$.

2.4 Prediction

After integration until steady state (criterion: $\|\dot{\mathbf{x}}^{(l)}\| < \epsilon \ \forall l$):

$$\hat{c} = \arg\max_c \sum_{l=0}^{L} \beta_l \sum_{m=1}^{M} \exp\left(-\frac{\|\mathbf{x}^{(l)}(T) - \boldsymbol{\mu}_{c,m}^{(l)}\|^2}{2(\sigma_{c,m}^{(l)})^2} \right) \tag{7}$$

where β_l are per-level weights (default: $\beta_l = 2^l$, learned or fixed).

2.5 Critical Design: Dimension-Dependent Kernel Initialisation (Condition I1)

Problem: in high dimension d, Euclidean distances concentrate around $\mathbb{E}[\|\mathbf{x}-\mathbf{y}\|] \approx \sqrt{d} \cdot \sigma_{\text{data}}$. With a fixed $\sigma_{\text{kernel}} \sim 1.0$, Gaussian kernels saturate toward 0 for $d \gtrsim 20$, forcing $\nabla E \to 0$ and halting learning.

Solution (I1): initialise with dimension scaling:

$$\sigma_{\text{init}}^{(l)} = \sqrt{d_l} \cdot \sigma_{\text{data}}^{(l)} \tag{8}$$

Empirical impact:

Dimension	Pre-fix kernel	Post-fix kernel	Gradient magnitude
10	0.95	0.88	0.24
100	0.03	0.67	0.18
784	< 0.01	0.52	0.15

Table 2: Kernel values and gradient magnitude by dimension (typical single run). Without I1, the framework fails to train beyond ∼20D in our experiments.

3 Learning via Gradient Descent

3.1 Three-Phase Progressive Training

To stabilise learning of coupled dynamics, we use a three-phase schedule:

Phase 1 (epochs 1–E_1): Independent learning. Set $\alpha_{\text{bu}} = \alpha_{\text{td}} = 0$. Each level self-organises independently. Ensures conditions C1 and C3 (see Section 4.1) are empirically satisfied before coupling is introduced.

Phase 2 (epochs E_1+1–E_1+E_2): Progressive coupling.

$$\alpha_{\text{bu}}(t)=\alpha_{\text{bu}}^{\max}\cdot\frac{t-E_1}{E_2},\quad \alpha_{\text{td}}(t)=\alpha_{\text{td}}^{\max}\cdot\frac{t-E_1}{E_2} \tag{9}$$

Phase 3 (epochs E_1+E_2+1–E_{total}): Full coupling. Fix $\alpha_{\text{bu}}=\alpha_{\text{bu}}^{\max}$, $\alpha_{\text{td}}=\alpha_{\text{td}}^{\max}$.

3.2 Composite Loss Function

$$\mathcal{L}=\mathcal{L}_{\text{CE}}+\mu_1\mathcal{L}_{\text{sep}}+\mu_2\mathcal{L}_{\text{dyn}} \tag{10}$$

Cross-entropy term (multi-level softmax):

$$\mathcal{L}_{\text{CE}}=-\frac{1}{N}\sum_{i=1}^{N}\log P(y_i\mid\{\mathbf{x}_i^{(l)}(T)\}_l) \tag{11}$$

Basin separation term:

$$\mathcal{L}_{\text{sep}}=\sum_{l}\sum_{c\neq c'}\sum_{m,m'}\exp\left(-\frac{\|\boldsymbol{\mu}_{c,m}^{(l)}-\boldsymbol{\mu}_{c',m'}^{(l)}\|^2}{2\sigma_{\min}^2}\right) \tag{12}$$

Dynamical regularisation:

$$\mathcal{L}_{\text{dyn}}=\frac{1}{T}\int_0^T\sum_l\|\dot{\mathbf{x}}^{(l)}(t)\|^2\,\mathrm{d}t \tag{13}$$

Gradients are computed via RK4 unrolling (for short T) or the adjoint method [1] via `torchdiffeq` (for long T, memory-efficient).

Attractor positions $\boldsymbol{\mu}_{c,m}^{(l)}$ are initialised by per-class k-means on the projected training data, providing an initial configuration that empirically satisfies the local-dominance assumption (C1) on simple problems.

3.3 Level-Aware Collapse Recovery

Proposition 3.1 (Level-Aware Recovery, informal mechanism)**.** *Define a level l^* as "collapsed" when its single-level accuracy $|\text{acc}_{l^*}-1/K|\leq\varepsilon$ for at least `patience` consecutive epochs during Phase 3, while global train accuracy is moderate (anti-false-positive filter). When detected, the following is triggered:*

1. *Re-anchor attractors of level l^* via supervised prototype initialisation in its latent space.*
2. *Temporarily attenuate top-down forces with cooldown (α_{td} scaled by $\tau<1$ for `td_cooldown_epochs`).*

This is a heuristic mechanism, not a theorem. Empirically on the concentric-rings benchmark (Campaign E8, 5 seeds 42–46), it provides a mean accuracy gain of +2.2 pts and a worst-seed gain of +4 pts. We treat this as engineering scaffolding, not as a contribution at the same epistemic level as the four theoretical propositions.

4 Theoretical Analysis

4.1 Convergence Guarantees

In the overdamped regime (large γ, negligible inertia), the system reduces to a pure gradient flow:

$$\dot{\mathbf{X}} = -\frac{1}{\gamma}\nabla_{\mathbf{X}}\mathcal{E}(\mathbf{X}) \tag{14}$$

where $\mathbf{X} = (\mathbf{x}^{(0)}, \ldots, \mathbf{x}^{(L)})$ and the total energy is:

$$\mathcal{E}(\mathbf{X}) = \sum_l E^{(l)}(\mathbf{x}^{(l)}) + \frac{\alpha_{\mathrm{bu}} + \alpha_{\mathrm{td}}}{2} \sum_l \|\mathbf{x}^{(l)} - \pi_l^{\dagger}(\mathbf{x}^{(l+1)})\|^2 \tag{15}$$

Proposition 4.1 (Dissipative Convergence). *$\mathcal{E}$ is bounded below by $-\sum_l \sum_{c,m} w_{c,m}^{(l)}$. Along any trajectory, $\dot{\mathcal{E}} = -\frac{1}{\gamma}\|\nabla\mathcal{E}\|^2 \leq 0$. By LaSalle's invariance principle, every bounded trajectory converges to the largest invariant subset of $\{\mathbf{X} : \nabla\mathcal{E}(\mathbf{X}) = 0\}$. The system can neither diverge nor oscillate indefinitely.*

Remark 4.2 (On the strength of Proposition 4.1). *Proposition 4.1 establishes convergence to* some *critical point of $\mathcal{E}$. It does not by itself guarantee convergence to a* minimum*, let alone to the* correct-class *minimum. The following result formalises sufficient conditions for the latter and is, we acknowledge, largely tautological in flavour: it states what every gradient method needs.*

Proposition 4.3 (Sufficient conditions for correct-basin convergence). *Assume:*

C1 – Local dominance *: at each level, the true-class attractor is the nearest attractor to the trajectory's neighbourhood in Euclidean distance.*

C2 – Sufficient coupling *: $\alpha_{\mathrm{bu}} + \alpha_{\mathrm{td}} > \alpha_{\mathrm{min}}$, a problem-dependent threshold determined by spectral analysis of the linearised system around fixed points.*

C3 – Basin separation *: $\mathcal{L}_{\mathrm{sep}}$ enforces a positive margin between attractors of different classes.*

Then any trajectory whose initial condition lies in the joint true-class basin converges to the true-class fixed point at each level.

Remark 4.4 (Honest framing)**.** *Proposition 4.3 is essentially:* if *the initial condition lies in the correct basin (C1)* and *the basins are separated (C3)* and *the coupling is well-tuned (C2),* then *gradient flow converges in that basin. We state it explicitly because (a) C1–C3 are precisely the conditions we monitor and enforce via k-means initialisation, $\mathcal{L}_{\text{sep}}$, and the three-phase schedule, and (b) the architectural value of HAML lies in making C1 hold at* higher *levels even when it fails at level 0, via top-down guidance (Proposition 4.5). The result is not deep in itself, but it makes the role of each design choice explicit.*

Proposition 4.5 (Hierarchical Guidance)**.** *Under coupling, the linearised dynamics around a fixed point have a block-tridiagonal Hessian (Eq. 16) whose minimum eigenvalue is bounded below by* $\min_l \lambda_{\min}(\mathbf{H}^{(l)}) + (\alpha_{\text{bu}} + \alpha_{\text{td}}) - 2\alpha_{\text{td}} \max_l \|\Pi_l\|_2$*. For PCA projections with* $\|\Pi_l\|_2 \leq 1$ *and moderate coupling, this lower bound is strictly positive, ensuring exponential local convergence at every level once the trajectory enters the basin of the higher level.*

The Hessian of $\mathcal{E}$ at a fixed point has block-tridiagonal structure:

$$\mathbf{H} = \begin{pmatrix} \mathbf{H}^{(0)} + \mathbf{C}^{(0)} & -\alpha_{\text{td}}\Pi_1^T & 0 & \cdots \\ -\alpha_{\text{td}}\Pi_1 & \mathbf{H}^{(1)} + \mathbf{C}^{(1)} & -\alpha_{\text{td}}\Pi_2^T & \cdots \\ 0 & -\alpha_{\text{td}}\Pi_2 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{16}$$

with $\mathbf{C}^{(l)} = (\alpha_{\text{bu}} + \alpha_{\text{td}})\mathbf{I}_{d_l}$. By Weyl's interlacing inequality [9], the coupling matrices $\mathbf{C}^{(l)}$ add a positive shift to the eigenvalues of the block-diagonal part; the off-diagonal $-\alpha_{\text{td}}\Pi_l$ blocks act as a perturbation bounded by $\alpha_{\text{td}}\|\Pi_l\|_2$. In the moderate-coupling regime $\alpha_{\text{td}}\|\Pi_l\|_2 < \alpha_{\text{bu}} + \alpha_{\text{td}}$, the coupling strictly accelerates local convergence.

4.2 Dimension-Independent Local Convergence Rate

Proposition 4.6 (Dimension-Independent Local Rate)**.** *At a fixed point* $\mathbf{x}^{(l)*} = \boldsymbol{\mu}_{c,m}^{(l)}$*, the intra-level Hessian block of* $E^{(l)}$ *in Eq. 2 reduces to:*

$$\mathbf{H}^{(l)}\big|_{\boldsymbol{\mu}} = \kappa^{(l)} \cdot \mathbf{I}_{d_l} \tag{17}$$

where

$$\kappa^{(l)} = \frac{w_{c,m}^{(l)}}{(\sigma_{c,m}^{(l)})^2} - \lambda \sum_{c' \neq c, m'} \frac{w_{c',m'}^{(l)}}{(\rho_{c',m'}^{(l)})^2} \exp\left(-\frac{\|\boldsymbol{\mu}_{c,m}^{(l)} - \boldsymbol{\mu}_{c',m'}^{(l)}\|^2}{2(\rho_{c',m'}^{(l)})^2}\right)$$

is a ***scalar independent of*** d_l*. The local linear convergence time* $T \sim \gamma/\kappa^{(l)}$ *does not grow with the level dimension.*

Remark 4.7 (Scope of Proposition 4.6). *This is a statement about the* local *linear rate at a fixed point. It does not address (a) the time to* enter *the basin, which depends on the energy landscape geometry away from the fixed point, nor (b) the global convergence time, which can still be dimension-sensitive. The result establishes that,* conditional on having reached the linear regime near an attractor*, the residual relaxation speed is set by a scalar that does not scale with d_l. Combined with condition I1 ($\sigma_{\text{init}} \propto \sqrt{d_l}$), this explains the empirical observation that the framework continues to train at 784D where naive constant-σ kernels fail (Table 2).*

Corollary 4.8. *Adding hierarchical levels does not, in itself, slow down local convergence at fixed points. By the eigenvalue interlacing analysis of Eq. 16, $\lambda_{\min}(\mathbf{H})$ remains bounded below by a quantity independent of L, provided $\alpha_{\text{td}} \max_l \|\Pi_l\|_2$ does not grow with L.*

4.3 Expressivity of the Hierarchy

Definition 4.9 (Topological complexity). *The topological complexity $\kappa(\mathcal{P})$ of a binary classification problem $\mathcal{P}$ is the minimum number of connected components of the boundary of a Bayes-optimal decision region.*

Proposition 4.10 (Hierarchical Expressivity Gain). *A flat ($L = 1$) attractor model with one Gaussian attractor per simply-connected basin requires at least $\kappa(\mathcal{P})$ attractors to represent a problem of topological complexity $\kappa(\mathcal{P})$. A hierarchical model with L levels and M attractors per level can represent up to $O(M^L)$ distinct joint configurations $(c, m_0, \ldots, m_L)$ with only $O(ML)$ parameters per class. On problems whose Bayes-optimal decision boundary factorises hierarchically (i.e. where a global discrete choice at level L selects a local geometry at level 0), the parameter count gap between flat and hierarchical is* exponential in L.

Remark 4.11 (Caveats). *Proposition 4.10 is an* existence-*style statement: it bounds the worst-case advantage of the hierarchy on a specific structural family of problems (hierarchically factorisable boundaries). It does not say that hierarchical models always beat flat ones — on problems without such factorisation, the gain may be zero. The concentric-rings problem (Section 5.3) is a constructive example where the factorisation holds: radial structure factorises out at level 1, angular refinement remains at level 0.*

Analytical example — alternating concentric rings: classes defined by $r \in [0, 1] \cup [2, 3]$ (class 1) and $r \in [1, 2] \cup [3, 4]$ (class 2). With $L = 1$ in the ambient 2D space, one needs $\Omega(1/\sigma)$ attractors per ring to cover its angular extent. With $L = 2$, level 1 (radial projection, $d_1 = 1$) resolves the global annular structure with 2 attractors per class; the top-down signal then constrains level 0 to refine locally.

Conjecture 4.12 (Necessity of Bidirectional Coupling). *We conjecture that there exist classification problems representable by a hierarchical bidirectional system with $O(ML)$ attractors that cannot be represented by L independent attractor levels with the same total attractor count. We do not provide a proof here. The empirical signature consistent with this conjecture is the qualitative recovery of non-convex decision boundaries by the coupled model on the concentric-rings benchmark (Section 5.3); but mean accuracy alone does not separate the two models statistically on our current sample size, so this remains conjectural.*

Proposition 4.13 (Rademacher Complexity Upper Bound). *For the coupled L-level classifier defined by Eq. 3 with K classes, M attractors per level, and n training samples, an upper bound on the empirical Rademacher complexity of the induced function class $\mathcal{F}_L$ takes the form:*

$$\mathcal{R}_n(\mathcal{F}_L) \leq c\sqrt{\frac{KML}{n}} \cdot \prod_{l=1}^{L}(1 + \alpha_l \|\Pi_l\|_F) \tag{18}$$

for a universal constant c depending on the Lipschitz constants of the integrated flow. The dependence on depth is $O(\sqrt{L})$ inside the square root and at most polynomial via the product term (which is bounded since $\|\Pi_l\|_F \leq \sqrt{d_l}$ and PCA spectral norm is bounded).

Remark 4.14. *The full proof requires a Lipschitz analysis of the flow map of Eq. 3; we sketch it in the supplementary material of the repository and acknowledge that tightening the constants is open work.*

4.4 Predictive Coding Correspondence and Bayesian Grounding

4.4.1 The Predictive Coding Generative Model

Predictive Coding [2,6] posits a hierarchical generative model:

$$p(\mathbf{x}^{(0:L)}) = p(\mathbf{x}^{(L)}) \prod_{l=0}^{L-1} p(\mathbf{x}^{(l)} \mid \mathbf{x}^{(l+1)}) \tag{19}$$

In the point-mean-field approximation, the variational free energy reduces to:

$$\mathcal{F}\big|_{\text{pt}} = \sum_{l} \frac{1}{2}\|\boldsymbol{\varepsilon}^{(l)}\|^2_{\Sigma^{(l)}} - \log p(\mathbf{x}^{(L)}) \tag{20}$$

where $\boldsymbol{\varepsilon}^{(l)} = \mathbf{x}^{(l)} - g^{(l+1)}(\mathbf{x}^{(l+1)})$ are prediction errors. The inference dynamics are:

$$\dot{\mathbf{x}}^{(l)} = -\frac{\partial \mathcal{F}}{\partial \mathbf{x}^{(l)}} = -\boldsymbol{\varepsilon}^{(l)} + \left(\frac{\partial g^{(l+1)}}{\partial \mathbf{x}^{(l)}}\right)^T \boldsymbol{\varepsilon}^{(l+1)} \tag{21}$$

This is the structural form of HAML's bidirectional coupling.

4.4.2 Term-by-Term Identification

With linear generators $g^{(l)}(\mathbf{x}^{(l)}) = \Pi_l^{\dagger}\mathbf{x}^{(l)}$ and isotropic covariances $\Sigma^{(l)} = \frac{1}{\alpha_{\mathrm{bu}}+\alpha_{\mathrm{td}}}\mathbf{I}$:

$$\mathcal{F}\big|_{\mathrm{linear}} = \frac{\alpha_{\mathrm{bu}}+\alpha_{\mathrm{td}}}{2}\sum_{l}\|\mathbf{x}^{(l)} - \Pi_l^{\dagger}\mathbf{x}^{(l+1)}\|^2 - \log p(\mathbf{x}^{(L)}) \tag{22}$$

The coupling terms are identical to those in $\mathcal{E}$ (Eq. 15). Identifying $E^{(l)}$ with $-\log p(\mathbf{x}^{(l)})$:

$$-E^{(l)}(\mathbf{x})\big|_{\lambda=0} = \log\sum_{c,m} w_{c,m}^{(l)}\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_{c,m}^{(l)}, (\sigma_{c,m}^{(l)})^2\mathbf{I}) + \mathrm{const} \tag{23}$$

Theorem 4.15 (MAP Equivalence under M1–M3)**.** *Under conditions:*

M1 – Linear generators *:* $g^{(l)}(\mathbf{x}^{(l)}) = \Pi_l^{\dagger}\mathbf{x}^{(l)}$ *(satisfied by construction in our architecture).*

M2 – Isotropic covariances *:* $\Sigma^{(l)} = \frac{1}{\alpha_{\mathrm{bu}}+\alpha_{\mathrm{td}}}\mathbf{I}$ *(satisfied when* α_{bu} *and* α_{td} *are level-independent; relaxable to per-level scalings).*

M3 – No repulsion *:* $\lambda = 0$ *in Eq. 2.*

With these three conditions, $\mathcal{E} \equiv \mathcal{F}|_{\mathrm{pt}}$ *as functionals of* $\mathbf{X}$ *(up to an additive constant), and HAML's gradient flow exactly performs* MAP inference *in a hierarchical generative model with Gaussian mixture prior at each level and Gaussian conditional likelihoods.*

With $\lambda > 0$*,* $\mathcal{E} = \mathcal{F}_{\mathrm{PC}} + \lambda\mathcal{F}_{\mathrm{discrim}}$*, where* $\mathcal{F}_{\mathrm{discrim}}$ *corresponds to a contrastive discriminative prior [4]. This hybrid is a discriminative augmentation of standard PC inference.*

The implicit generative model is:

$$p(\mathbf{x}^{(l)}) = \sum_{c,m} w_{c,m}^{(l)}\mathcal{N}(\mathbf{x}^{(l)} \mid \boldsymbol{\mu}_{c,m}^{(l)}, (\sigma_{c,m}^{(l)})^2\mathbf{I}) \tag{24}$$

$$p(\mathbf{x}^{(l-1)} \mid \mathbf{x}^{(l)}) = \mathcal{N}\Big(\mathbf{x}^{(l-1)}\ \Big|\ \Pi_l^{\dagger}\mathbf{x}^{(l)},\ \tfrac{1}{\alpha_{\mathrm{bu}}+\alpha_{\mathrm{td}}}\mathbf{I}\Big) \tag{25}$$

This Bayesian grounding suggests a natural VAE extension, replacing the point-mean-field approximation with an ELBO objective (Section 7.3).

5 Empirical Validation

We conduct six campaigns across synthetic and real datasets. We report results with their statistical status (single-seed vs. multi-seed, with means ± standard deviations where available) and resist drawing conclusions that are not supported by the sample sizes.

5.1 Missing Baselines: A Limitation Acknowledged Upfront

We acknowledge that the empirical study does *not* include strong neural baselines (MLP, CNN) trained under identical protocols. On the datasets we use, established baselines achieve approximately:

- **MNIST (full)**: logistic regression $\sim 92\%$, MLP $\sim 98\%$, CNN $> 99\%$.
- **MNIST subset** (1500 train / 500 test, 5 epochs): a 2-layer MLP typically reaches 90%+ in seconds.
- **Fashion-MNIST (5000 train)**: logistic regression $\sim 84\%$, 2-layer MLP $\sim 88\%$, small CNN $> 90\%$.

Our 75.8% on MNIST-subset and 80.1% on Fashion-MNIST are therefore *below standard baselines*. We do not present HAML as a more accurate classifier on these datasets. The empirical claims of this preprint concern (i) the framework trains stably in high dimension, (ii) the coupling provides a measurable benefit on structurally non-convex problems (rings, intermediate-noise moons), and (iii) the attractor geometry is interpretable and inspectable. Reaching baseline-competitive accuracy is left as explicit future work.

5.2 Campaign A: MNIST Subset — Gradient Training Gains and Coupling Neutrality

Protocol: 1500 train / 500 test from MNIST, levels $784 \to 392 \to 196$, 2 attractors per class, 5 epochs, 3-phase training, seed 42.

Condition	Test Accuracy	Phase Progress (train)
k-means init. only (no gradient training)	45.6%	—
End of Phase 1 (independent levels)	—	64.2%
End of Phase 2 (progressive coupling)	—	68.7%
End of Phase 3 (full coupling)	—	71.8%–77.8%
Full training, independent variant ($\alpha = 0$)	75.8%	—
Full training, coupled ($\alpha_{\text{bu}} = \alpha_{\text{td}} = 1$)	75.8%	—

Table 3: Campaign A. Gradient training raises accuracy from 45.6% (k-means initialisation alone) to 75.8% (+30.2 pts). On this short, near-convex protocol, **bidirectional coupling provides no measurable benefit** over the independent variant. Single seed; the +30.2 pts gain compares to an extremely weak baseline (k-means class assignment with no gradient learning) and should not be interpreted as evidence of competitiveness with standard MNIST classifiers (see Section 5.1).

Interpretation. This is the most important negative result of the study: on a nearly-convex 10-class problem, coupling is neutral. This is consistent with the theory (Proposition 4.10 requires high topological complexity to predict an advantage). We treat this result as informative for scoping the regime in which coupling helps, not as evidence against the architecture.

5.3 Campaign B: Concentric Rings — Coupling on Non-Convex Geometry

Protocol: alternating concentric rings (topological complexity $\kappa \geq 2$), $n_{\text{runs}} = 5$, seeds 42–46.

Model	**Accuracy (mean ± std)**	**Qualitative boundary**
Independent levels ($\alpha = 0$)	$67.30\% \pm 4.37$	Convex (fails annular topology)
Coupled (tuned)	$71.20\% \pm 6.43$	Non-convex (aligns rings)
Δ (mean)	+3.90 pts	—

Table 4: Campaign B. The mean gain (+3.90 pts) is smaller than the pooled standard deviation ($\sigma_{\text{coupled}} = 6.43$), so a Welch's two-sample t-test on $n = 5$ seeds per group does not reject the null hypothesis of equal means at standard significance levels. However, the qualitative shift in the learned decision boundary (visible in the per-seed plots in the repository) is consistent across all five coupled runs.

Honest reading. The quantitative mean difference is not statistically significant at $n = 5$ seeds. The qualitative geometric shift (convex decision boundary in the independent variant; ring-aligned non-convex boundary in the coupled variant) is robust across all five seeds and is the actual evidence in favour of the coupling on this benchmark. The high variance in the coupled condition motivates Campaign E and the level-aware recovery mechanism.

5.4 Campaign C: Noisy Moons — Noise Robustness Profile (Single Seed)

Protocol: `make_moons` with noise $\sigma \in \{0.10, 0.20, 0.30, 0.40\}$, $n = 3000$, **single seed (42)**.

Falsifiable prediction. If the bell-shaped profile is real and not a single-seed artefact, multi-seed runs (10+ seeds) should reproduce: (a) peak coupling gain in the intermediate-noise regime, (b) no significant gain at very low or very high noise. We state this as a falsifiable prediction the reader can test against the released code.

Noise	HAML-Indep.	HAML-Coupled	KNN	SVM RBF
0.10	99.60%	99.60%	99.87%	99.87%
0.20	89.33%	**95.60%**	96.53%	97.20%
0.30	88.13%	90.53%	90.80%	91.33%
0.40	84.67%	85.07%	83.33%	87.33%

Table 5: Campaign C. **Single seed.** The coupling delta (HAML-Coupled − HAML-Indep.) is $0.00, 6.27, 2.40, 0.40$ pts at noise $0.10, 0.20, 0.30, 0.40$. The maximum at intermediate noise is consistent with the prediction that top-down context helps most when local signal is ambiguous but exploitable. Multi-seed replication is required before treating the bell-shaped profile as established.

5.5 Campaign D: HAML vs. a Controlled SA-nODE Reimplementation

Important caveat. The SA-nODE comparison uses a *controlled internal reimplementation* of the architecture described in [5], not the authors' official code, which we did not run for this study. The reimplementation reproduces: single ambient space, binary attractors $\pm a$, analytic double-well potential, linear coupling layer trained jointly with the dynamics, grid search over $(a, \Delta t, \gamma)$. A reviewer is justified in regarding this as a weak baseline: a paper comparison should ultimately use the official code. We report the result as preliminary.

Noise	HAML Coupled	SA-nODE (reimpl.)	Delta
0.10	**99.60%**	86.67%	+12.93 pts
0.20	**95.60%**	86.40%	+9.20 pts
0.30	**90.53%**	85.87%	+4.67 pts
0.40	**85.07%**	83.33%	+1.73 pts

Table 6: Campaign D. **Internal SA-nODE reimplementation, single seed.** The hierarchical architecture dominates our reimplementation of the flat single-space model across noise levels. SA-nODE degrades more slowly in absolute terms (−3.33 pts vs. −14.53 pts for HAML across the noise range), which we do not currently explain. We flag this comparison as preliminary pending evaluation against the official SA-nODE code.

5.6 Campaign E: Stabilisation and Inter-Level Collapse

We iteratively develop the level-aware collapse guard (Section 3.3) over subtests E1–E10. Full per-subtest results are in Appendix A.

Subtest	Mechanism	Mean Acc.	Min (worst seed)
E8 baseline (coupled)	None	$68.60\% \pm 7.54$	56.67%
E8 with level-aware guard	Sec. 3.3	$70.80\% \pm 5.54$	60.67%
E10 ($n = 3200$, seeds 42–46)	Cause-oriented patch	$78.75\% \pm 7.63$	68.25%

Table 7: Campaign E summary. The level-aware guard reduces variance and improves the worst-seed performance on the concentric-rings protocol. Scaling sample size from 1200 to 3200 (E10) lifts the mean accuracy by ≈ 8 pts.

Mechanistic finding: collapsed levels show `separation_ratio` ≈ 0.15 (low) despite moderate compactness — suggesting a learning blockade rather than total geometric collapse. Re-anchoring of the affected level's attractors resolves the blockade within 1–2 epochs.

Cross-dataset validation (mutex baseline, seeds 42–46, $n = 3200$):

- `noisy_moons` (noise 0.30): $89.13\% \pm 1.73$, min 86.25%.
- `make_classification` (4 classes): $74.08\% \pm 4.31$, min 67.12%.

5.7 Campaign F: Fashion-MNIST — Stable Training in 784D (Single Seed)

Protocol: Fashion-MNIST (784D, 10 classes), $n_{\text{train}} = 5000$, $n_{\text{test}} = 1000$, levels $784 \to 392 \to 196$, 10 epochs, phases $(2, 3, 5)$, **single seed (42)**.

Metric	Value
Test accuracy	**80.1%**
Train accuracy (final)	82.3%
End of Phase 1	75.36%
End of Phase 2	78.86%
End of Phase 3	82.26%
Level-aware guard events	0 (naturally stable run)
Compute (Colab GPU)	6 979 s (≈2 hrs)
Reproducibility (Kaggle GPU)	80.1% / 6 222 s

Table 8: Campaign F. **Single seed.** The headline result is that the three-phase training progresses smoothly through all phases in 784D without collapse, not that 80.1% is a strong accuracy on Fashion-MNIST (it is not; see Section 5.1). Identical results across two GPU environments confirm deterministic behaviour.

Compute caveat. At ≈2 hrs per 10-epoch run, a multi-seed Fashion-MNIST campaign was beyond the compute budget available for this preprint.

Multi-seed validation is the most important empirical follow-up.

6 Mechanistic Diagnostics

We implement diagnostic tools to isolate failure modes:

6.1 Level-Wise Accuracy Tracking

At each epoch, per-level accuracy $\mathrm{acc}_l(t)$ is logged. Divergence $\Delta_l = \max_l \mathrm{acc}_l - \min_l \mathrm{acc}_l$ serves as a collapse indicator. Typical seed-42 signature on the concentric-rings task:

- Phase 1 (independent): all levels converge to $\sim 60\%$.
- Phase 2 (progressive): divergence emerges ($\Delta_l \approx 0.15$).
- Phase 3 (full coupling): one level plateaus at ~ 0.5; guard triggers re-anchoring.

6.2 Attractor Geometry Metrics

Per level:

- `intra_spread_mean`: within-class attractor spread (compactness).
- `inter_centroid_dist_min`: minimum between-class centroid distance.
- `separation_ratio`: geometric margin proxy.

Collapsed levels show `separation_ratio` ≈ 0.15 (low) despite non-zero compactness, indicating a learning blockade rather than total geometric collapse.

6.3 Force Distribution Analysis

Histograms of $\|\mathbf{F}^{(l)}\|$ per level reveal:

- *Saturated kernels* (pre-fix I1 in high dimensions): $\|\mathbf{F}\| \approx 0$.
- *Balanced learning* (post-fix I1): approximately Gaussian distribution.
- *Gradient death* in late phases: indicator of convergence or instability.

7 Discussion

7.1 What the Evidence Supports, and What It Does Not

Evidence supports:

1. The framework trains stably in dimensions up to 784, given condition I1.
2. On structurally non-convex problems (concentric rings), coupled HAML *qualitatively* recovers ring-aligned decision boundaries that the independent variant fails to produce.
3. Bidirectional coupling provides a single-seed accuracy gain at intermediate noise on noisy moons (consistent with theory; awaiting multi-seed confirmation).
4. Exact MAP correspondence with hierarchical Predictive Coding under explicit conditions M1–M3.

Evidence does not support:

1. HAML being competitive with standard neural baselines on MNIST or Fashion-MNIST on accuracy alone.
2. A statistically significant mean accuracy gain from coupling at $n = 5$ seeds on rings.
3. Superiority over SA-nODE based on the authors' official implementation.
4. Scalability beyond 784D.

7.2 Genuine Limitations

1. **Missing strong baselines**: no MLP, CNN, or boosted-tree baseline trained under identical protocols. This is the most important methodological gap.
2. **Single-seed runs on real data**: MNIST-subset and Fashion-MNIST are single-seed for compute reasons.
3. **Statistical underpowering on rings**: $n = 5$ seeds is insufficient to claim a mean accuracy advantage from coupling; we rely on qualitative geometric evidence.
4. **Internal SA-nODE reimplementation**: not a substitute for evaluation against the authors' official code.

5. **Computational cost**: ≈2 hrs per 10-epoch run on Fashion-MNIST (5000 samples). Adjoint methods and gradient checkpointing should reduce this substantially.
6. **Inter-level instability**: variance remains high between seeds; the level-aware guard helps but does not fully resolve it.
7. **Hyperparameter sensitivity**: $(\alpha_{\text{bu}}, \alpha_{\text{td}}, \mu_1, \mu_2)$ require per-dataset tuning. Automated search is an open engineering challenge.
8. **Unexplained asymmetry**: SA-nODE-reimpl. degrades more slowly than HAML as noise increases (Campaign D); we do not currently have a mechanistic explanation.
9. **Conjecture status**: Proposition 4.12 (bidirectional necessity) is stated as a conjecture without a proof.

7.3 Future Work

1. **Strong baselines**: MLP / CNN / boosted trees under identical protocols on all reported datasets.
2. **Multi-seed Fashion-MNIST**: at least 5 seeds with the adjoint method to amortise compute.
3. **Official SA-nODE comparison**: replication against Marino et al.'s released code.
4. **Adjoint integration**: full neural-ODE adjoint sensitivity to reduce memory and training time.
5. **Generative extension (HAML-VAE)**: replace the point-mean-field approximation with an ELBO objective.
6. **Proof of Conjecture 4.12**: a constructive separation between bidirectional and independent-levels representations.
7. **Tighter Rademacher constants** and sample complexity bounds.
8. **Scalability beyond 784D**: tested on CIFAR-10 (3072D) and larger.

8 Reproducibility and Implementation

All code is available at: `https://github.com/ikobootloader/HIERARCHICAL-ATTRACTOR-MACHINE-LEARNING`

8.1 Key Software

- `haml/`: main package (dynamics, model, training, projection, analysis, visualisation).
- `experiments/`: 6 reproducible campaign scripts (A–F).
- `diagnose_forces.py`: high-dimension saturation detector.
- Dependencies: `numpy, scikit-learn, torch, torchdiffeq, scipy`.

8.2 Reproducible Commands

Campaign A (MNIST ablation):

```
python experiments/mnist_coupling_ablation.py
```

Campaign B (concentric rings, multi-seed):

```
python experiments/concentric_coupling_ablation.py --n-runs 5 --save-figure
```

Campaign C (noisy benchmark):

```
python experiments/noisy_benchmark.py
```

Campaign F (Fashion-MNIST, GPU):

```
PYTHONIOENCODING=utf-8 python experiments/fashion_mnist_short_probe.py \
  --device cuda --seed 42 --n-train 5000 --n-test 1000 --n-epochs 10 \
  --phase1-epochs 2 --phase2-epochs 3
```

9 Conclusion

We introduced HAML, a hierarchical bidirectional attractor framework for supervised classification, combining PCA-decomposed state spaces, learned continuous attractors, and Predictive-Coding-style coupling. The core architectural contribution is the integration of these three elements within a single dissipative dynamical system, together with a practical training recipe that empirically stabilises it.

Theoretical contributions comprise: dissipative convergence (Proposition 4.1), dimension-independent local rate at fixed points (Proposition 4.6), a hierarchical expressivity argument (Propositions 4.10–4.13), and an exact correspondence with hierarchical Predictive Coding MAP inference under explicit conditions (Theorem 4.15).

Empirically, the evidence is preliminary and mixed. The framework trains stably in 784D (80.1% Fashion-MNIST, single seed, below standard baselines). Coupling provides a qualitative geometric advantage on structurally non-convex problems (rings) and a single-seed accuracy gain at intermediate noise (moons), but no measurable gain on near-convex problems

(MNIST subset). Multi-seed validation and comparison against strong neural baselines remain the most important next steps.

We release the framework and all experimental code in the hope that it will be useful as a starting point for further investigation of attractor-based classification, hierarchical Predictive Coding implementations, and energy-based hybrids.

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A Detailed Campaign Results

A.1 Campaign E — Subtest Summary

Subtest	Key mechanism	Accuracy (test)	Δ vs. indep.
E1	Reinforced coupling ($\alpha_{\text{td}} = 1.5$, 25 ep.)	86.38%	+15.63 pts
E2	Early-stop + LR decay	82.88%	+12.00 pts
E3	Adaptive μ_{sep} + soft landing ep. 20	69.88%	−1.25 pts
E4	Soft landing on divergence trigger	76.25%	+6.37 pts
E5	Raised trigger + patience 2	76.13%	+5.00 pts
E6	Collapse guard on divergence jump	86.38%	+16.00 pts
E7	Seed-wise diagnostics + LR cascade patch	69.33% $\pm$ 7.10	+9.60 pts
E8	Level-aware guard (phase 3 only)	70.80% $\pm$ 5.54	+11.07 pts
E10	Scale $n = 3200$, cause-oriented patch	78.75% $\pm$ 7.63	—

Table 9: Stabilisation iteration summary (Campaign E). Results on the concentric-rings protocol, except E10 where $n = 3200$, seeds 42–46. The high single-seed peaks (E1, E6) were not retained because their stability across seeds is not established. E8 was selected as the operational mechanism on the multi-seed mean–variance trade-off.

A.2 Initialisation Condition I1 — Justification

Distance concentration in $\mathbb{R}^d$:

$$\mathbb{E}[\|\mathbf{x} - \mathbf{y}\|] \approx \sqrt{d} \cdot \sigma_{\text{data}} \tag{26}$$

For a Gaussian kernel $\exp(-r^2/2\sigma^2)$ to maintain activations in the range $[0.1, 0.9]$ across dimension d, σ must scale as $\sqrt{d}$. Without I1, the following table shows the resulting kernel saturation (single typical run).

Dimension	Pre-fix kernel	Post-fix kernel	Gradient
10	0.95	0.88	0.24
100	0.03	0.67	0.18
784	< 0.01	0.52	0.15

Table 10: Kernel activations and gradient magnitudes with and without condition I1.

B Hyperparameter Sensitivity

Ablation on the concentric-rings protocol:

Parameter	Range tested	Accuracy range
α_{bu}	$[0.0, 1.0]$	$[67.3\%, 73.2\%]$
α_{td}	$[0.0, 1.5]$	$[67.3\%, 75.1\%]$
μ_1 (separation)	$[0.1, 1.0]$	$[70.5\%, 71.9\%]$
μ_2 (dynamics)	$[0.001, 0.1]$	$[71.1\%, 71.3\%]$

Table 11: Sensitivity on primary hyperparameters. The model is moderately robust; α_{bu} and α_{td} require the most attention per dataset.

C Summary of Theoretical Results

Result	Section	Informal statement
Prop. 4.1 (Convergence)	4.1	Bounded trajectories converge to critical points; no divergence or sustained oscillation.
Prop. 4.3 (Correct basin)	4.1	Under C1–C3, gradient flow stays in the correct basin (conditional, not deep).
Prop. 4.5 (Hierarchical guide)	4.1	Eigenvalue analysis shows coupling accelerates local convergence.
Prop. 4.6 (Dim. independence)	4.2	Local rate at a fixed point is dimension-independent.
Prop. 4.10 (Expressivity)	4.3	Exponential parameter gain on hierarchically factorisable problems.
Conj. 4.12 (Bidi. necessity)	4.3	Bidirectional coupling is conjectured necessary on certain problems (no proof).
Prop. 4.13 (Rademacher)	4.3	Sub-linear-in-depth complexity bound.
Thm. 4.15 (MAP equivalence)	4.4	Exact MAP in hierarchical Gaussian-mixture model under M1–M3.
Prop. 3.1 (Level-aware recovery)	3.3	Heuristic engineering mechanism reduces inter-seed variance.

Table 12: Summary of theoretical results, with their epistemic status.